Task 4: Sentiment Analysis

Abstract

This project performs sentiment analysis on customer reviews using Python and the TextBlob NLP library. Reviews are classified into Positive, Negative, or Neutral sentiments.

Objectives

• Analyze customer reviews.
• Classify sentiments.
• Visualize sentiment distribution.
• Generate useful business insights.

Tools Used

Google Colab, Python, Pandas, TextBlob, Matplotlib, Microsoft Excel

Dataset

A dataset of 20 customer reviews stored in reviews.xlsx was used.

Methodology

1. Created the dataset.
2. Uploaded it to Google Colab.
3. Imported libraries.
4. Read the Excel file.
5. Applied TextBlob sentiment analysis.
6. Generated sentiment labels.
7. Created a bar chart.
8. Exported the final dataset.

Python Code

1. import pandas as pd

from textblob import TextBlob

2. from google.colab import files

uploaded = files.upload()

3. df = pd.read_excel("reviews.xlsx")

df.head()

4. def sentiment(text):

```
    score = TextBlob(text).sentiment.polarity
```

```
    if score > 0:
        return "Positive"
    elif score < 0:
        return "Negative"
    else:
        return "Neutral"
```

5. df["Sentiment"] = df["Review Text"].apply(sentiment)

Df

6. import matplotlib.pyplot as plt

```
# Count sentiments
sentiment_counts = df["Sentiment"].value_counts()

# Create bar chart
sentiment_counts.plot(kind="bar")

# Add title and labels
plt.title("Sentiment Analysis")
plt.xlabel("Sentiment")
plt.ylabel("Number of Reviews")

# Show chart
plt.show()
```

7. df.to_excel("Sentiment_Result.xlsx", index=False)

Output Screenshots

Insert:

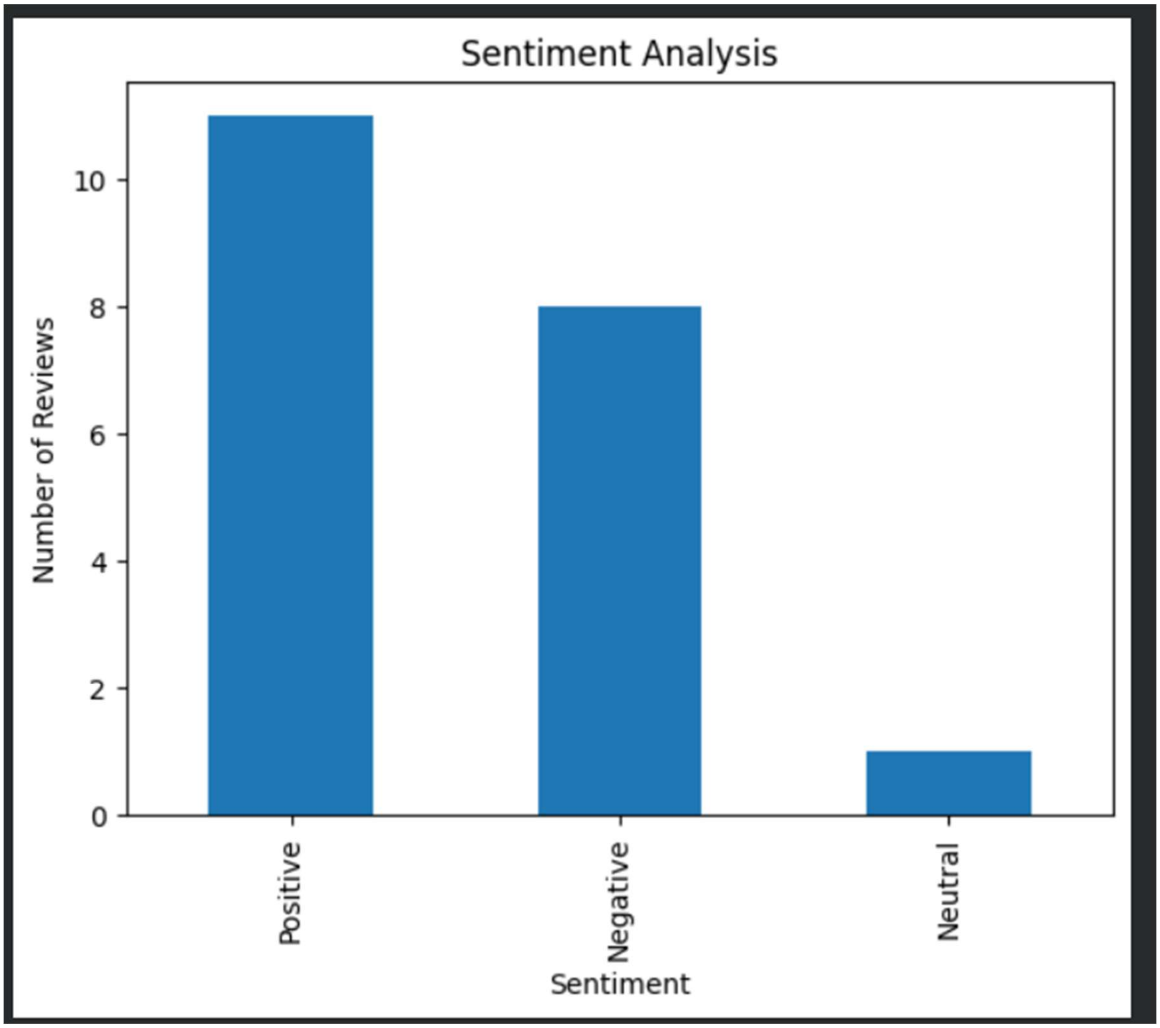
1. Dataset loaded (df.head())

	Review ID	Review Text
0	1	This product is amazing.
1	2	I love this phone.
2	3	Very bad quality.
3	4	Delivery was late.
4	5	Excellent customer service.

2. Sentiment output table

	Review ID	Review Text	Sentiment
0	1	This product is amazing.	Positive
1	2	I love this phone.	Positive
2	3	Very bad quality.	Negative
3	4	Delivery was late.	Negative
4	5	Excellent customer service.	Positive
5	6	Average product.	Negative
6	7	Waste of money.	Negative
7	8	Highly recommended.	Positive
8	9	Not worth buying.	Negative
9	10	Good value for money.	Positive
10	11	Fantastic experience.	Positive
11	12	Packaging was damaged.	Neutral
12	13	Very satisfied.	Positive
13	14	Poor battery life.	Negative
14	15	It's okay.	Positive
15	16	Fast delivery.	Positive
16	17	Terrible support.	Negative
17	18	Awesome quality.	Positive
18	19	I am disappointed.	Negative
19	20	Best purchase ever.	Positive

3. Bar chart



4. Downloaded Excel output

```
from google.colab import files
files.download("Sentiment_Result.xlsx")
```

```
Downloading "Sentiment_Result.xlsx":
```

Result

The analysis successfully classified reviews into Positive, Negative, and Neutral categories and displayed the distribution using a bar chart.

Conclusion

Sentiment analysis provides valuable insights into customer opinions and helps organizations improve products, services, and marketing strategies.

References

1. https://pandas.pydata.org
2. https://textblob.readthedocs.io
3. https://matplotlib.org